



VYSOKÉ UČENÍ FAKULTA
TECHNICKÉ INFORMAČNÍCH
V BRNĚ TECHNOLOGIÍ



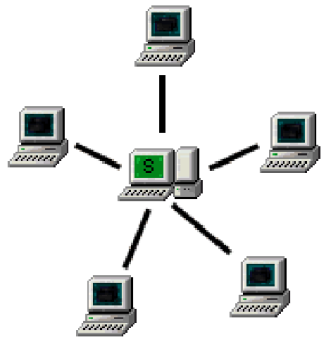
9

Rekapitulace

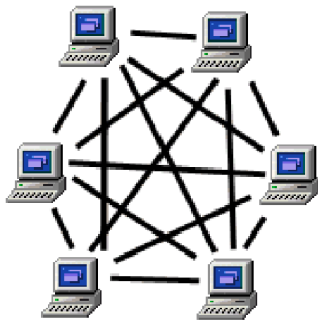
IPK 2021 L

Typ komunikace

Server Based Network



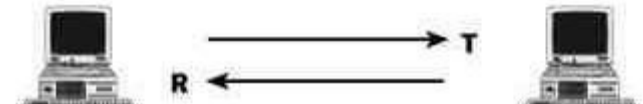
Peer to Peer Network



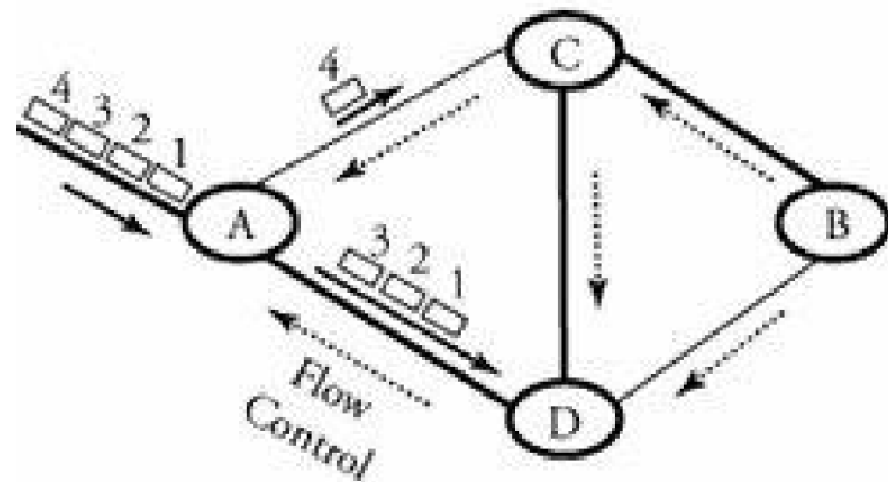
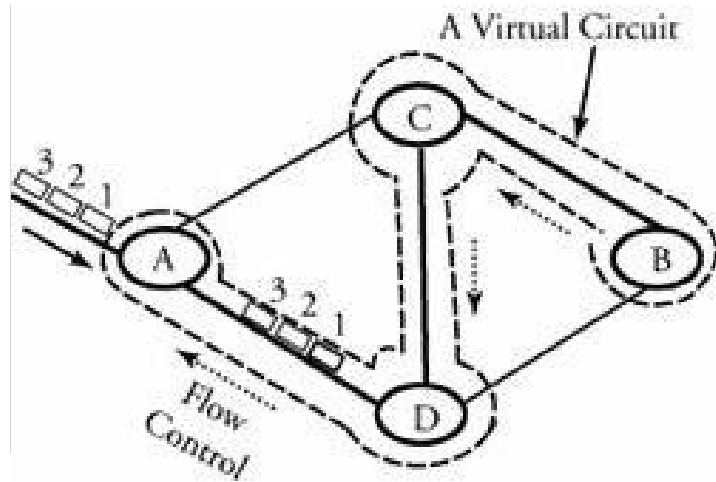
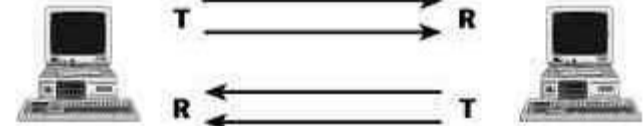
Simplex



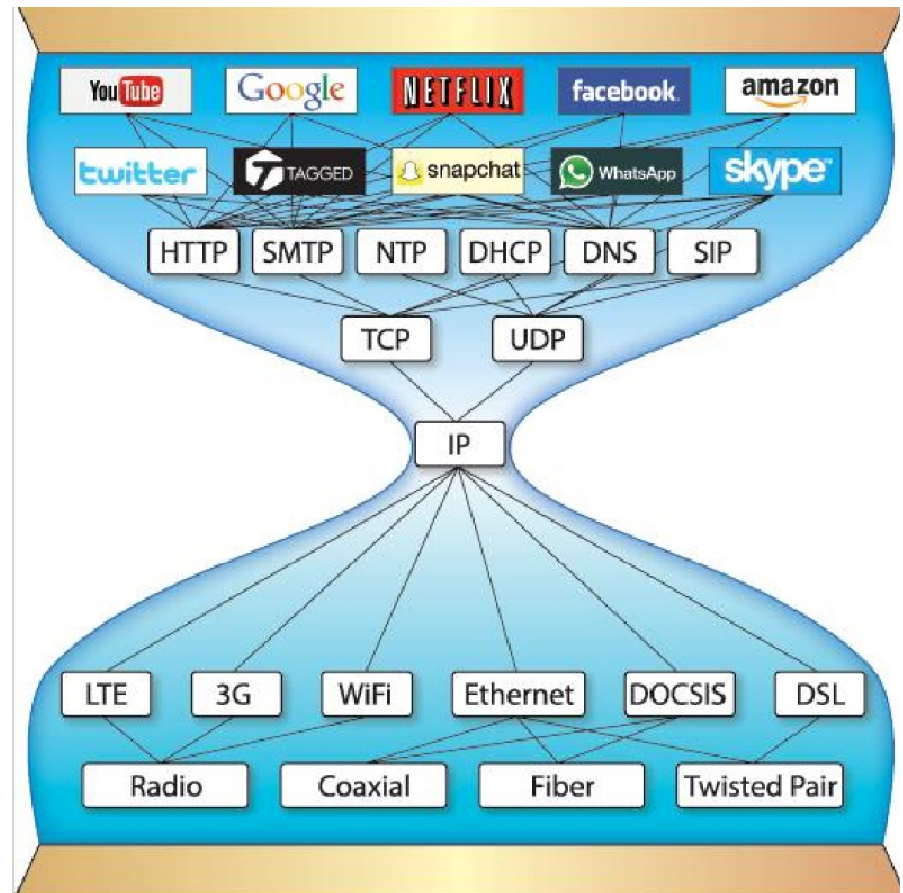
Half-Duplex
(2-Wire Circuit)



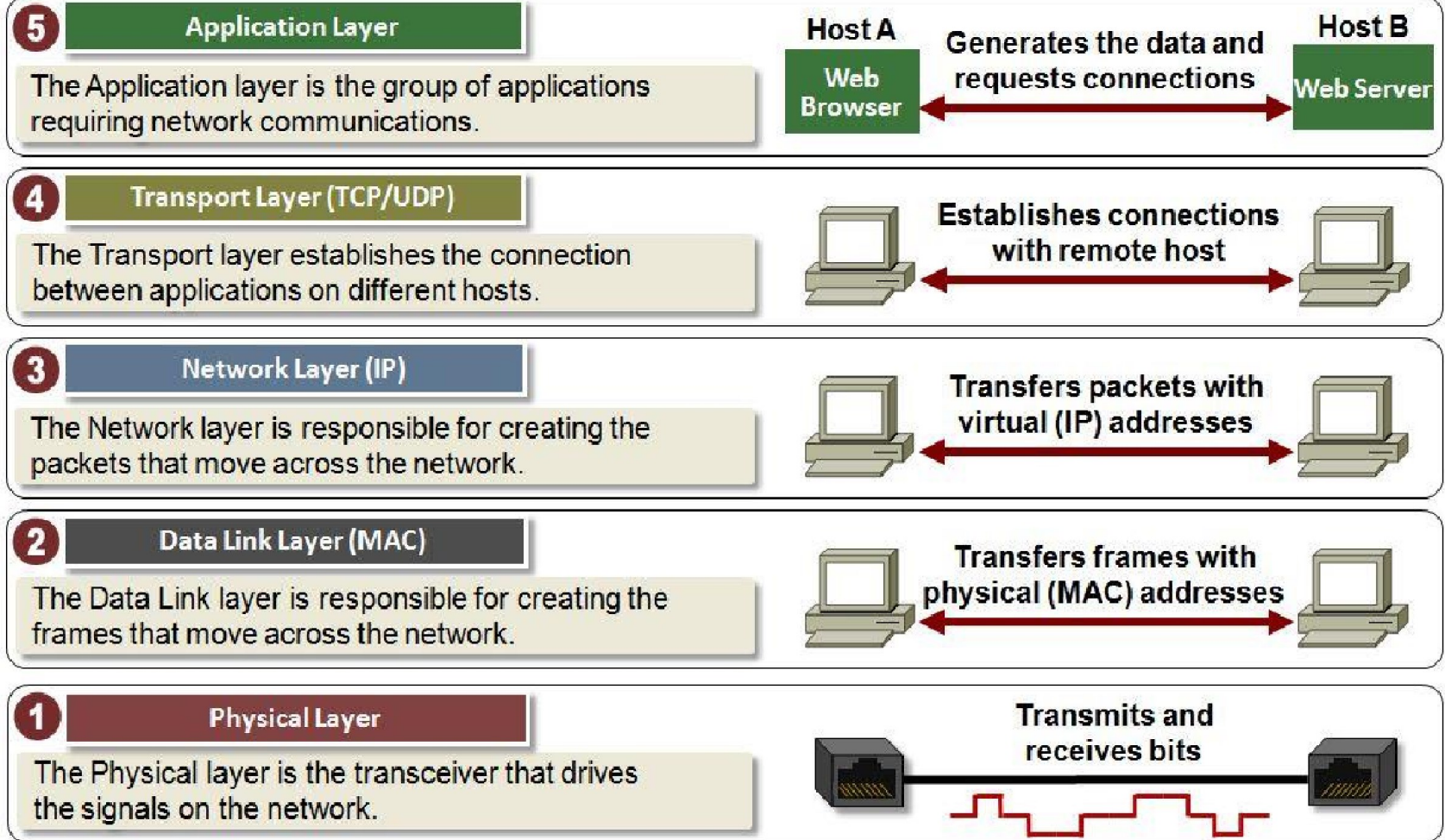
Full-Duplex
(4-Wire Circuit)



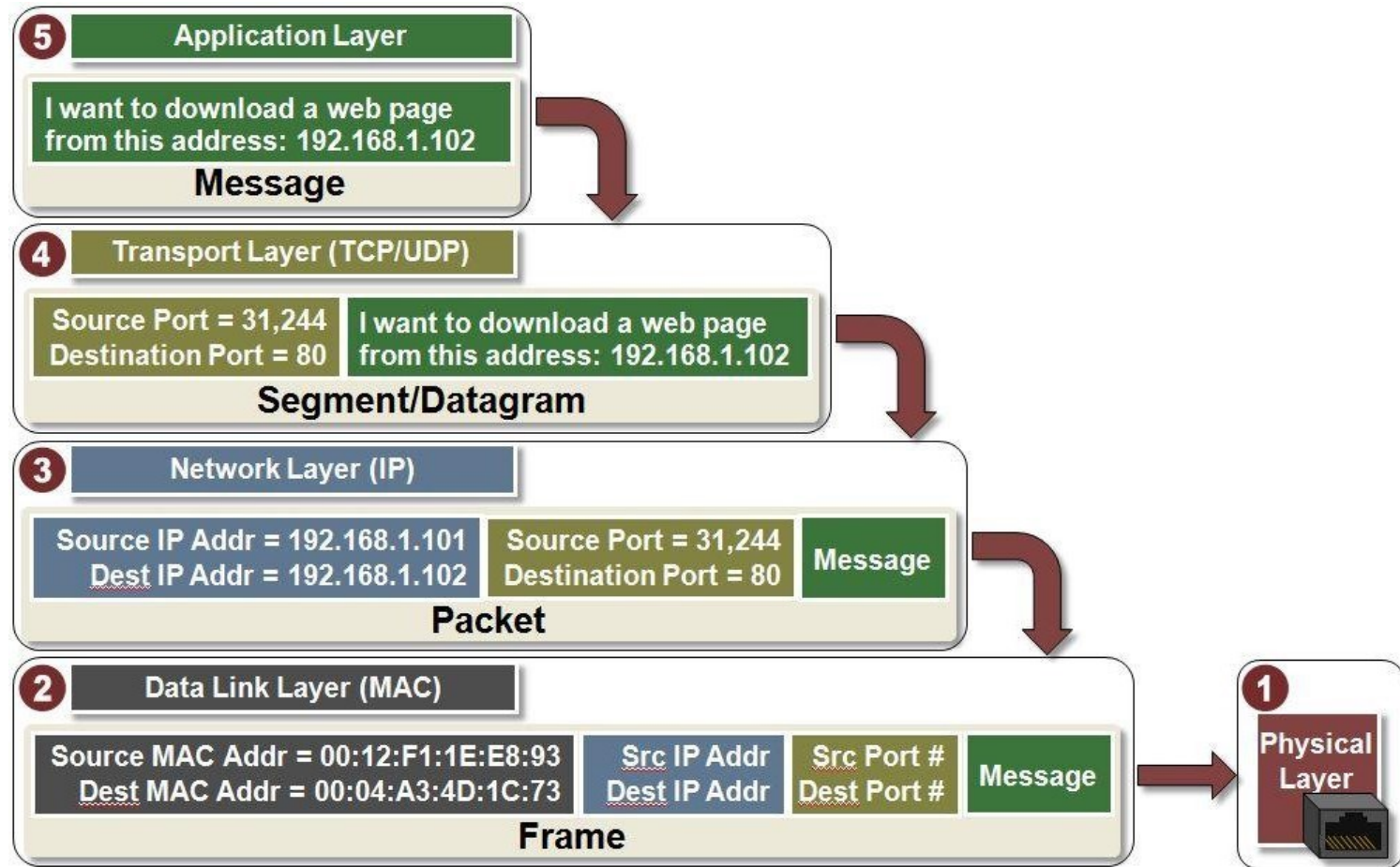
Protokol



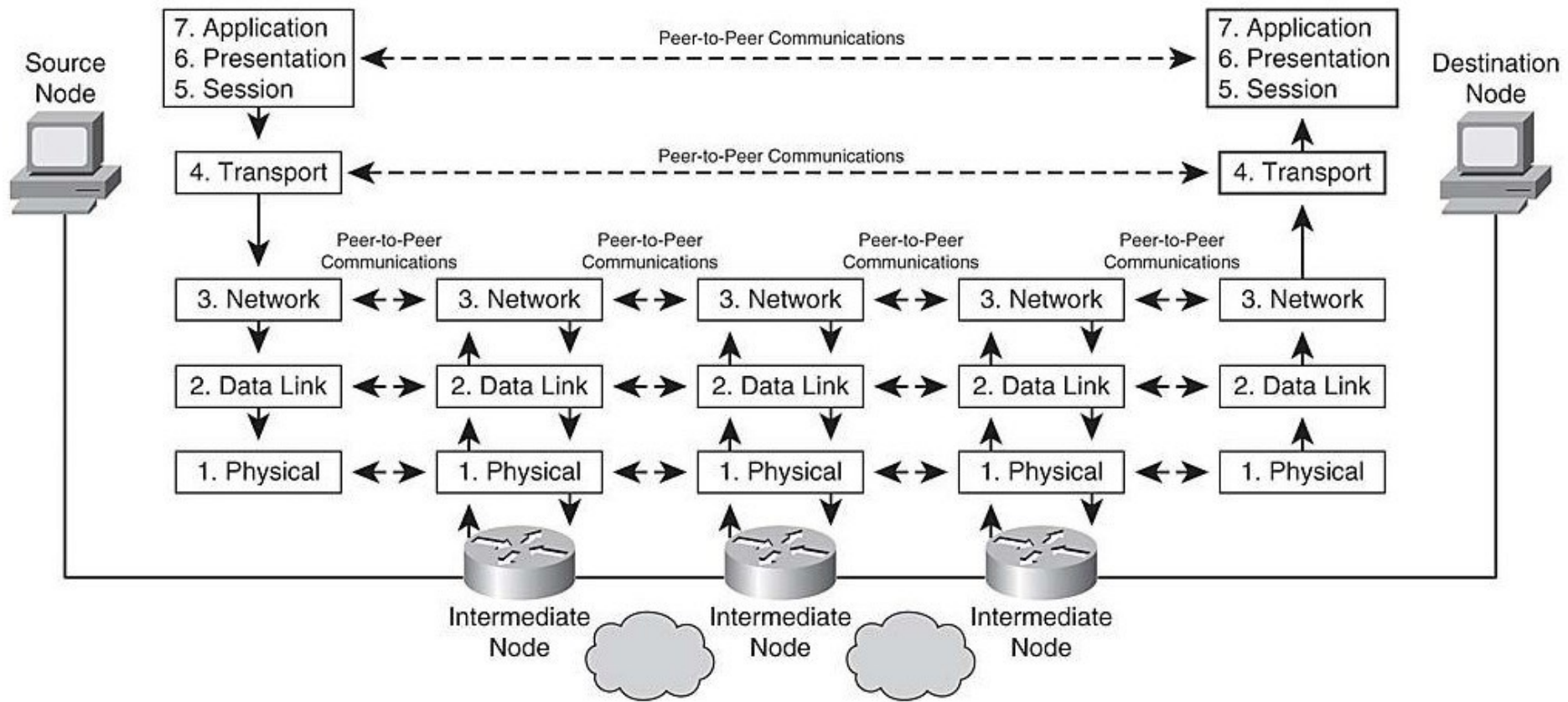
Protokolový zásobník



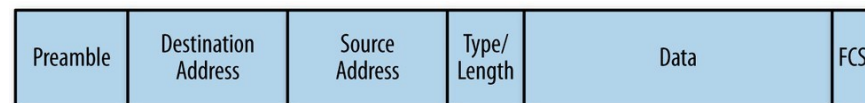
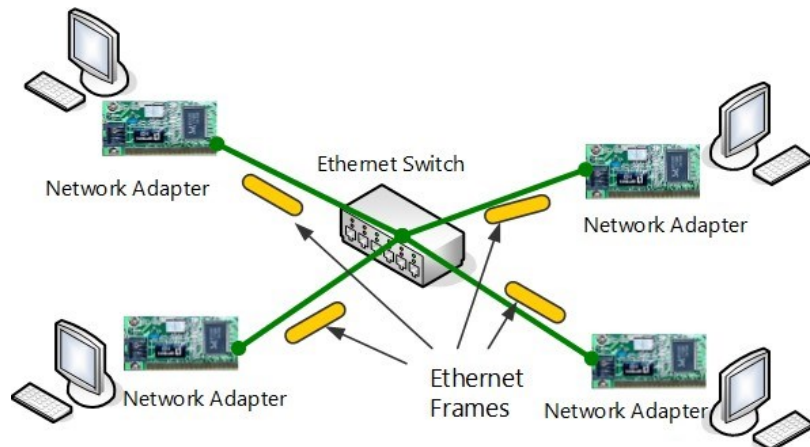
Encapsulate



Komunikace mezi koncovými stanicemi



Zařízení- SWITCH



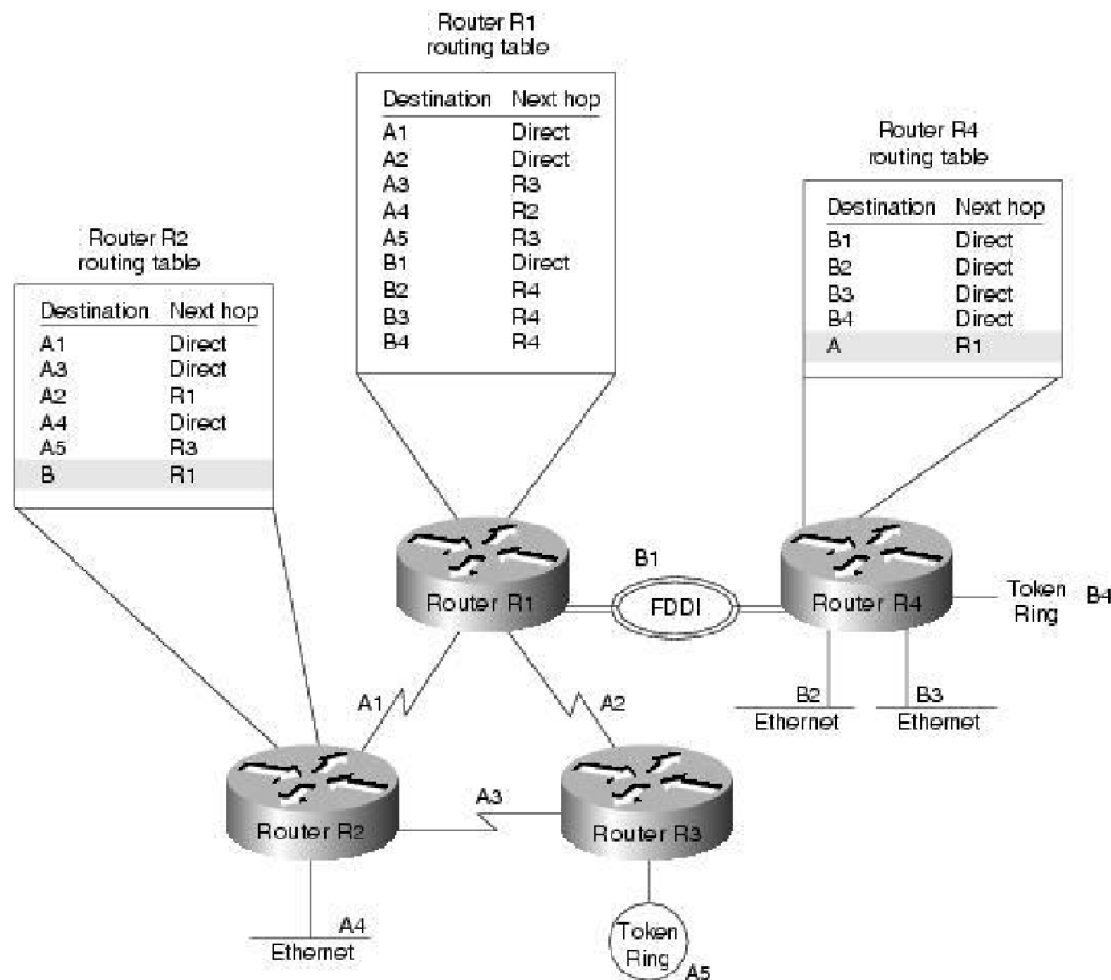
Ethernet Frame

```
Switch#show mac-address-table
```

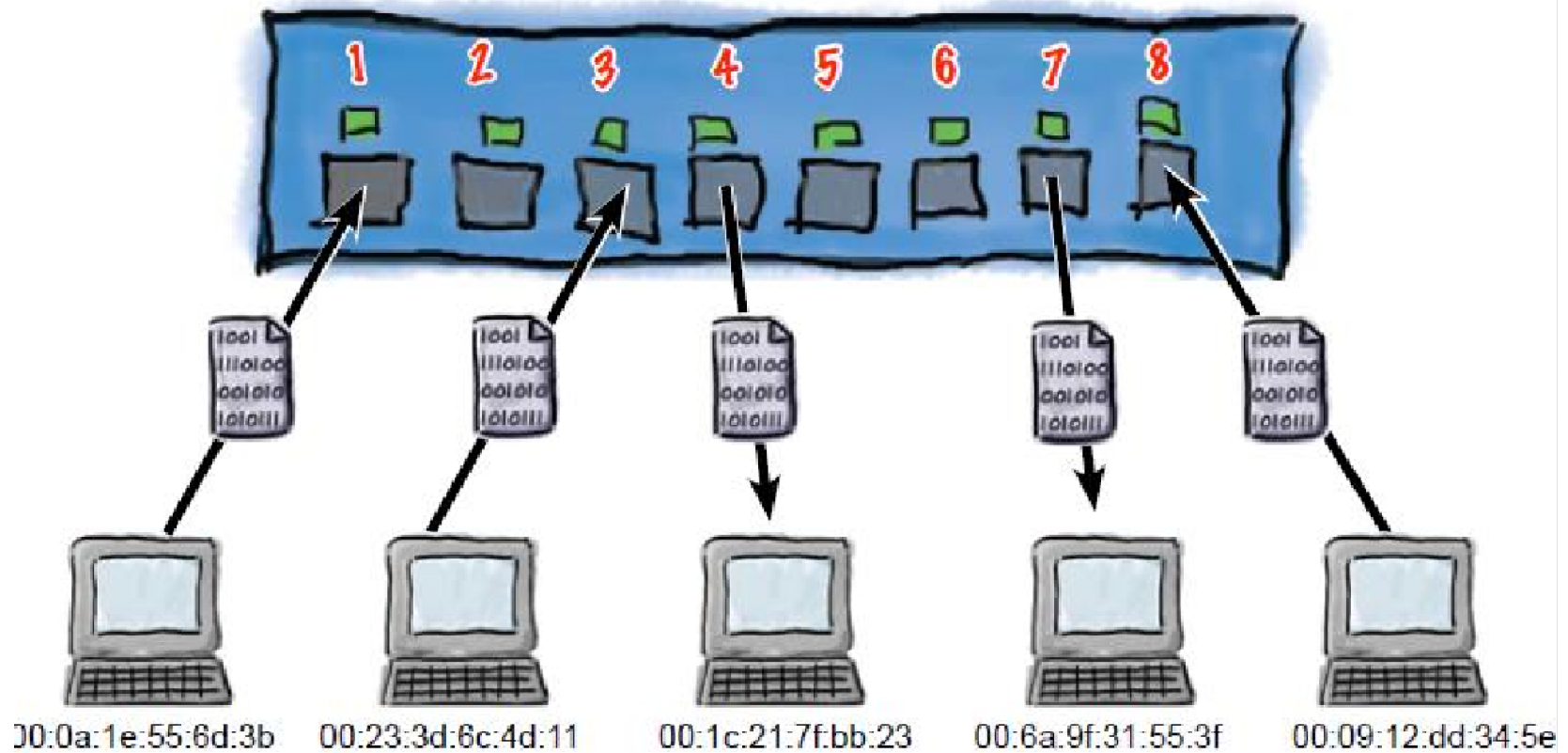
Mac Address Table

Mac Address	Ports
800C.E799.D8CH	1
9A7Y.F100.1HAV	2
A0E1.Y212.M2DA	3
108H.1NI9.HG11	4

Směrovač

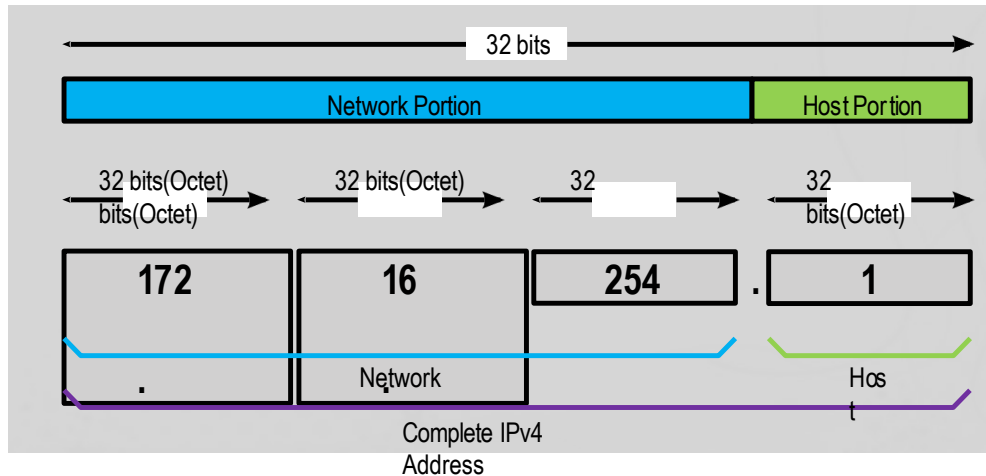


Adresování - MAC



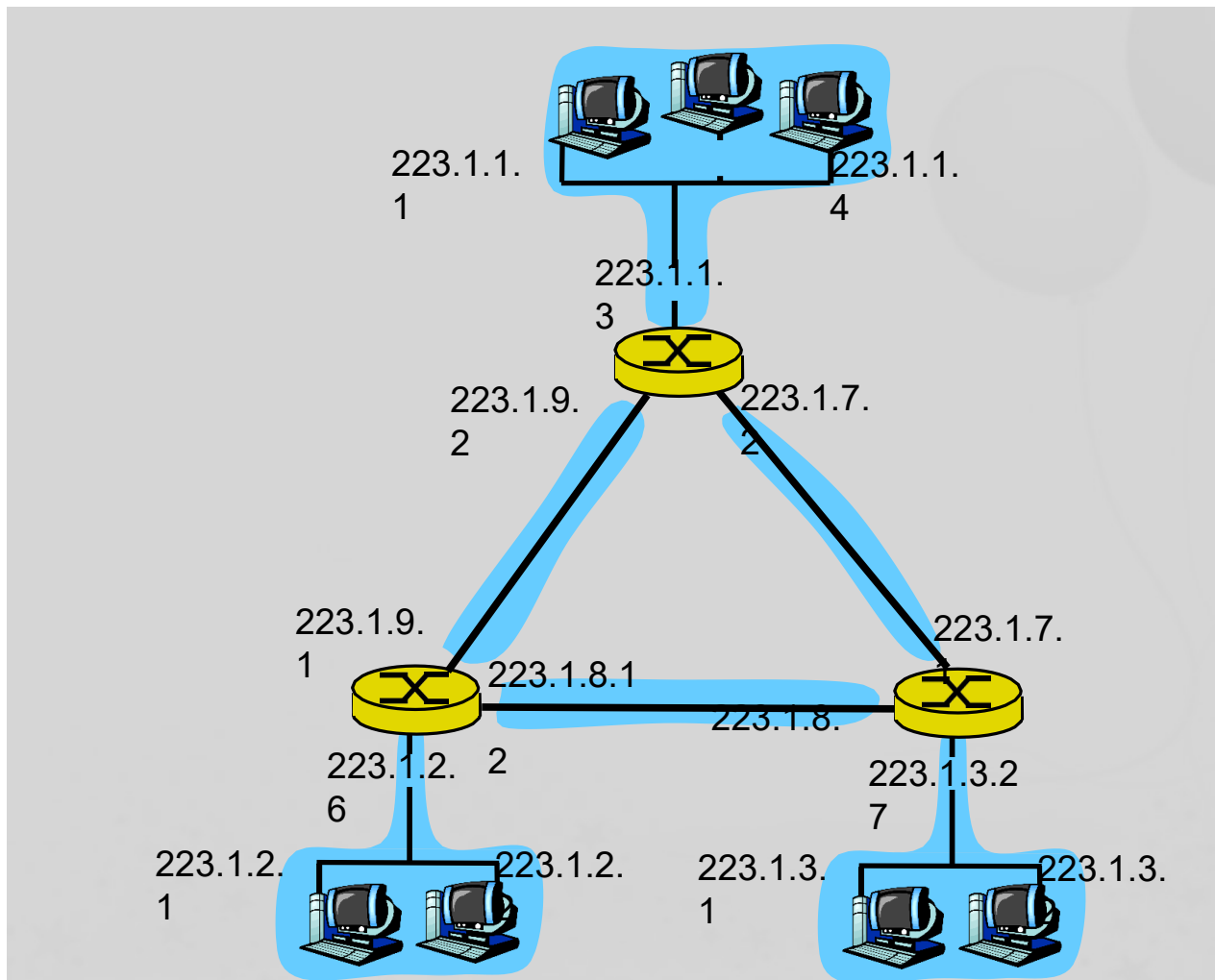
nes.
.fi
t

Adresování IPv4



Class	Leading bits	Size of network number bit field	Size of rest bit field	Number of networks	Addresses per network	Total addresses in class	Start address	End address
Class A	0	8	24	128 (2^7)	16,777,216 (2^{24})	2,147,483,648 (2^{31})	0.0.0.0	127.255.255.255
Class B	10	16	16	16,384 (2^{14})	65,536 (2^{16})	1,073,741,824 (2^{30})	128.0.0.0	191.255.255.255
Class C	110	24	8	2,097,152 (2^{21})	256 (2^8)	536,870,912 (2^{29})	192.0.0.0	223.255.255.255
Class D (multicast)	1110	not defined	not defined	not defined	not defined	268,435,456 (2^{28})	224.0.0.0	239.255.255.255
Class E (reserved)	1111	not defined	not defined	not defined	not defined	268,435,456 (2^{28})	240.0.0.0	255.255.255.255

Adresování sítí



Subnetting

IP address 192.35.128.93



Binary Form

Subnet Host bits

11000000.00100011.10000000.01011101

Extended network prefix

11111111.11111111.11111111.11100000

Subnet Mask

Address Range Net-1 (192.168.10.128/25)

Network Address

192.168.10.10000000 = 192.168.10.128

First Usable IP Address for Host

192.168.10.10000001 = 192.168.10.129

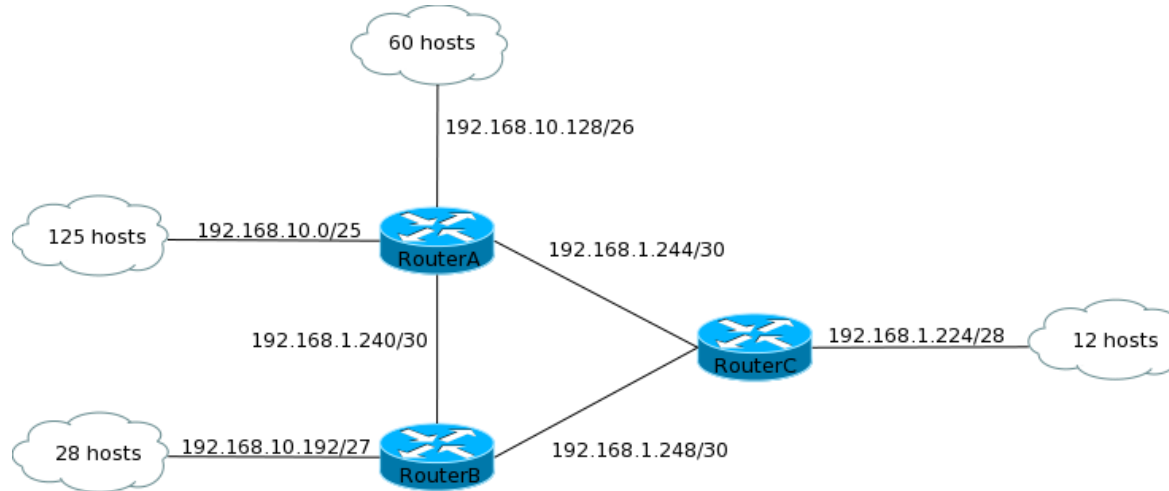
Last Usable IP Address for Host

192.168.10.11111110 = 192.168.10.254

Broadcast IP Address

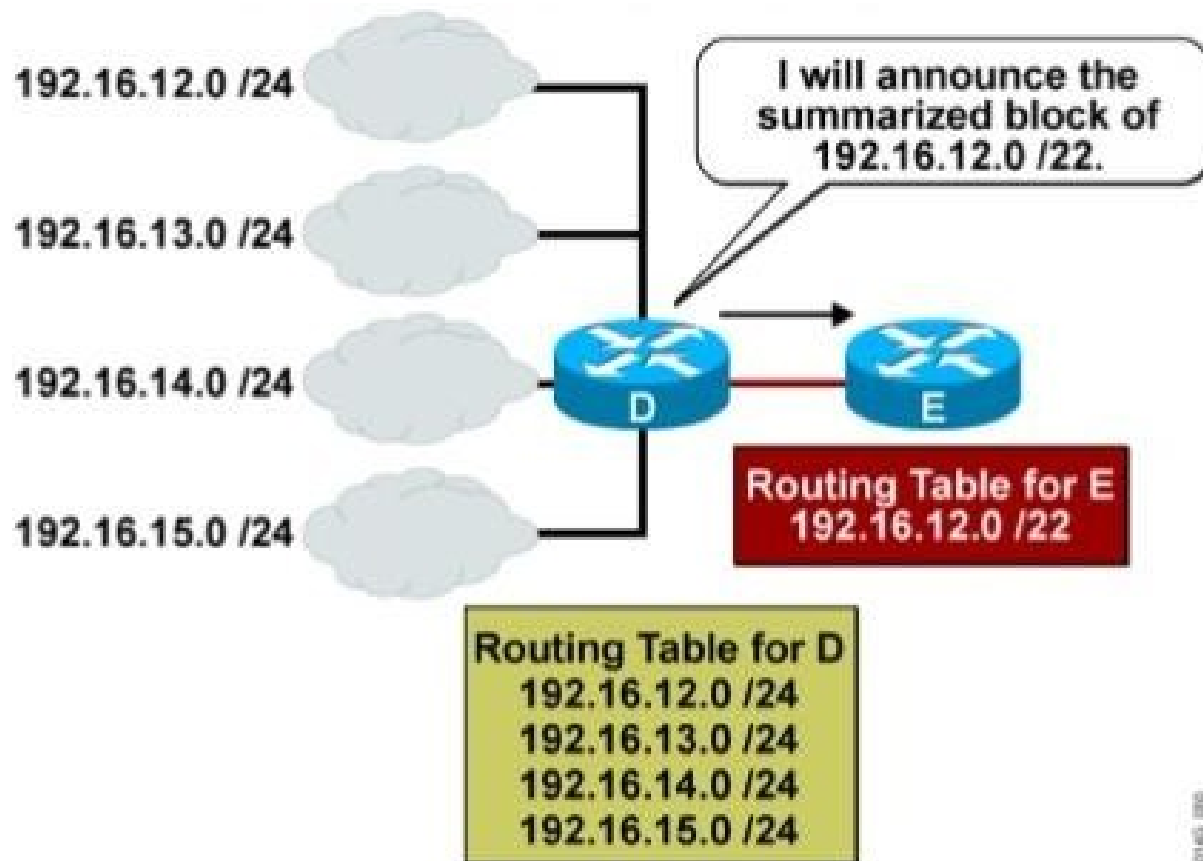
192.168.10.11111111 = 192.168.10.255

VLSM

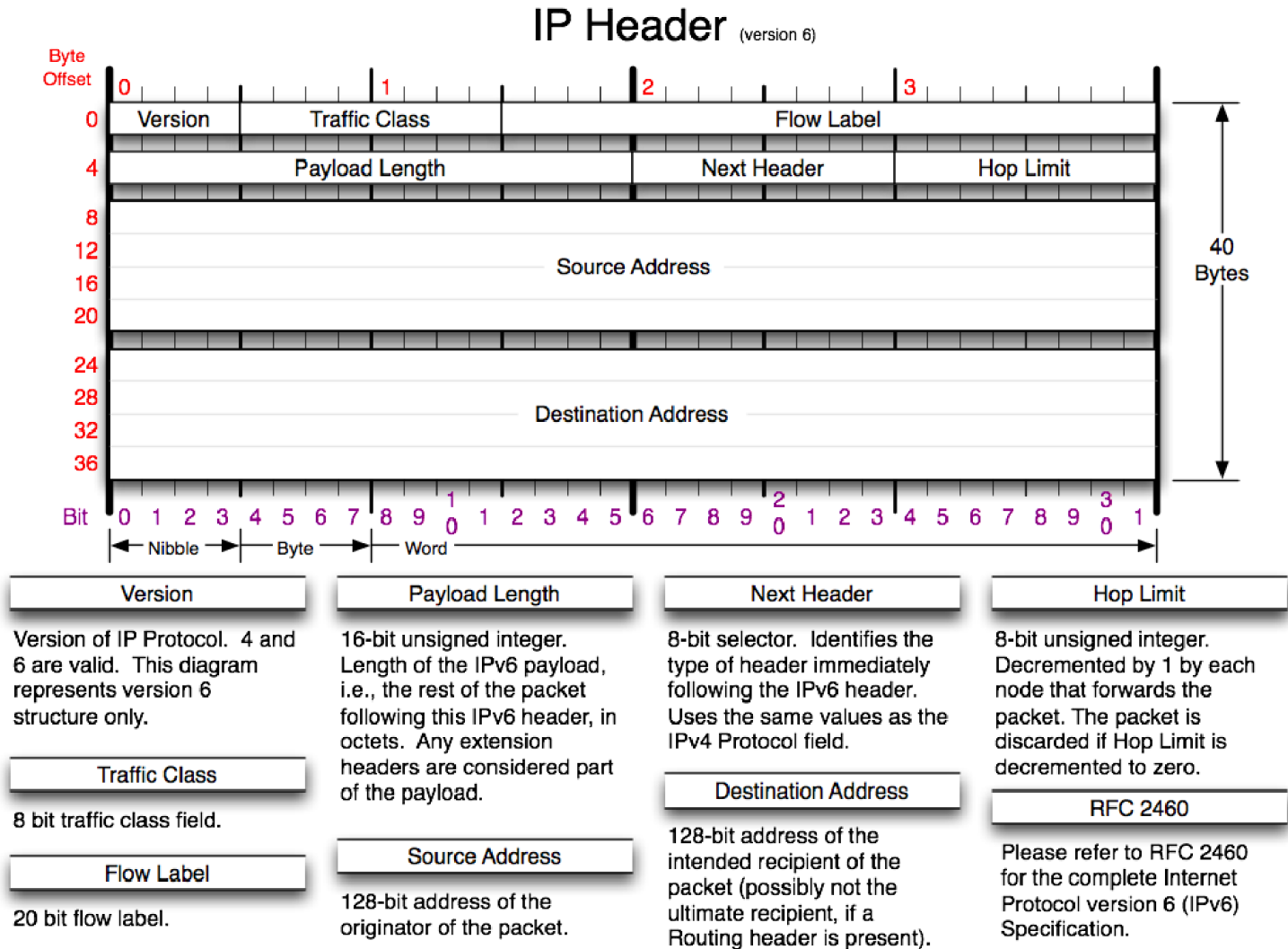


Subnet Prefix / CIDR	Subnet mask	Usable IP address/hosts	Usable IP addresses + Network and Broadcast address
/26	255.255.255.192	62	64
/27	255.255.255.224	30	32
/28	255.255.255.240	14	16
/29	255.255.255.248	6	8
/30	255.255.255.252	2	4

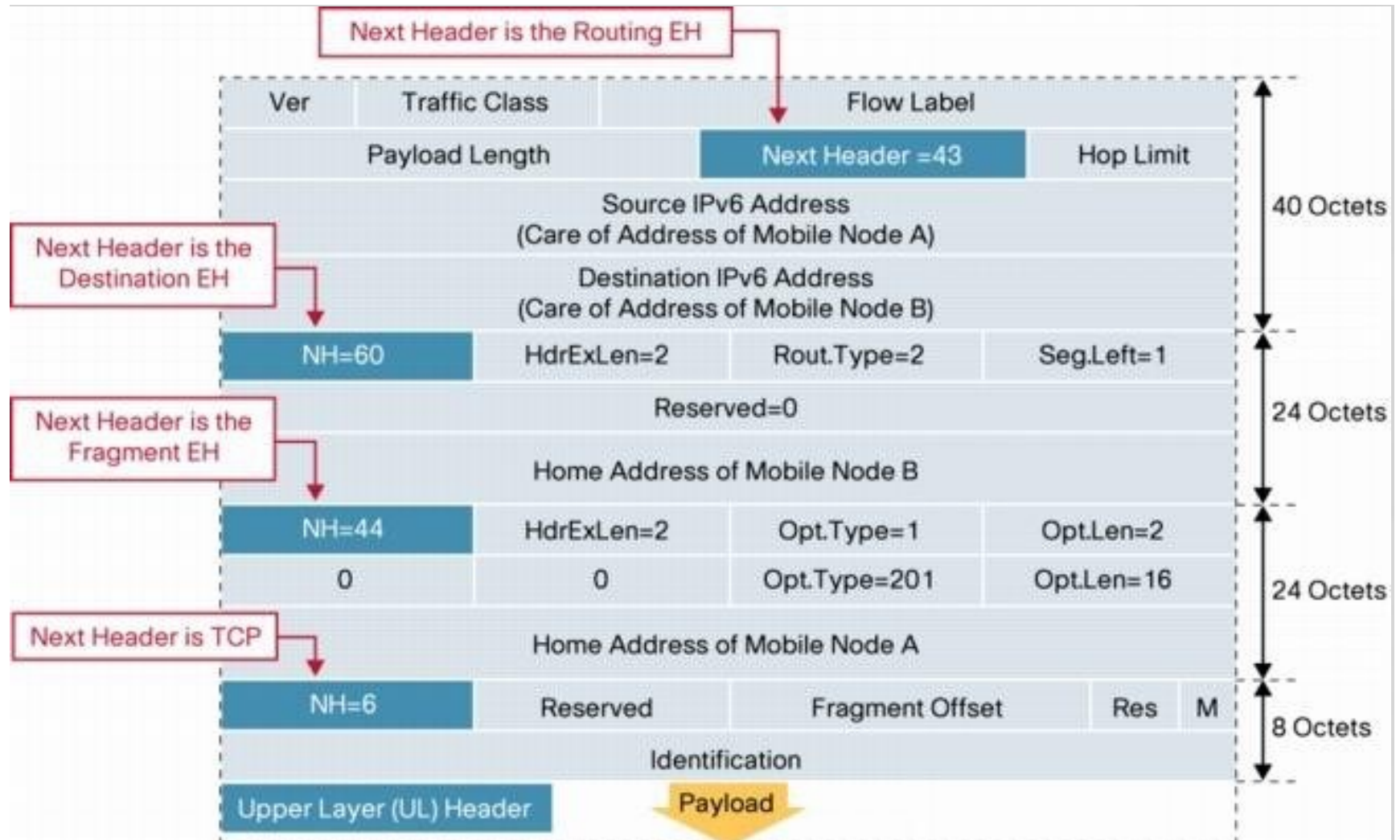
CIDR



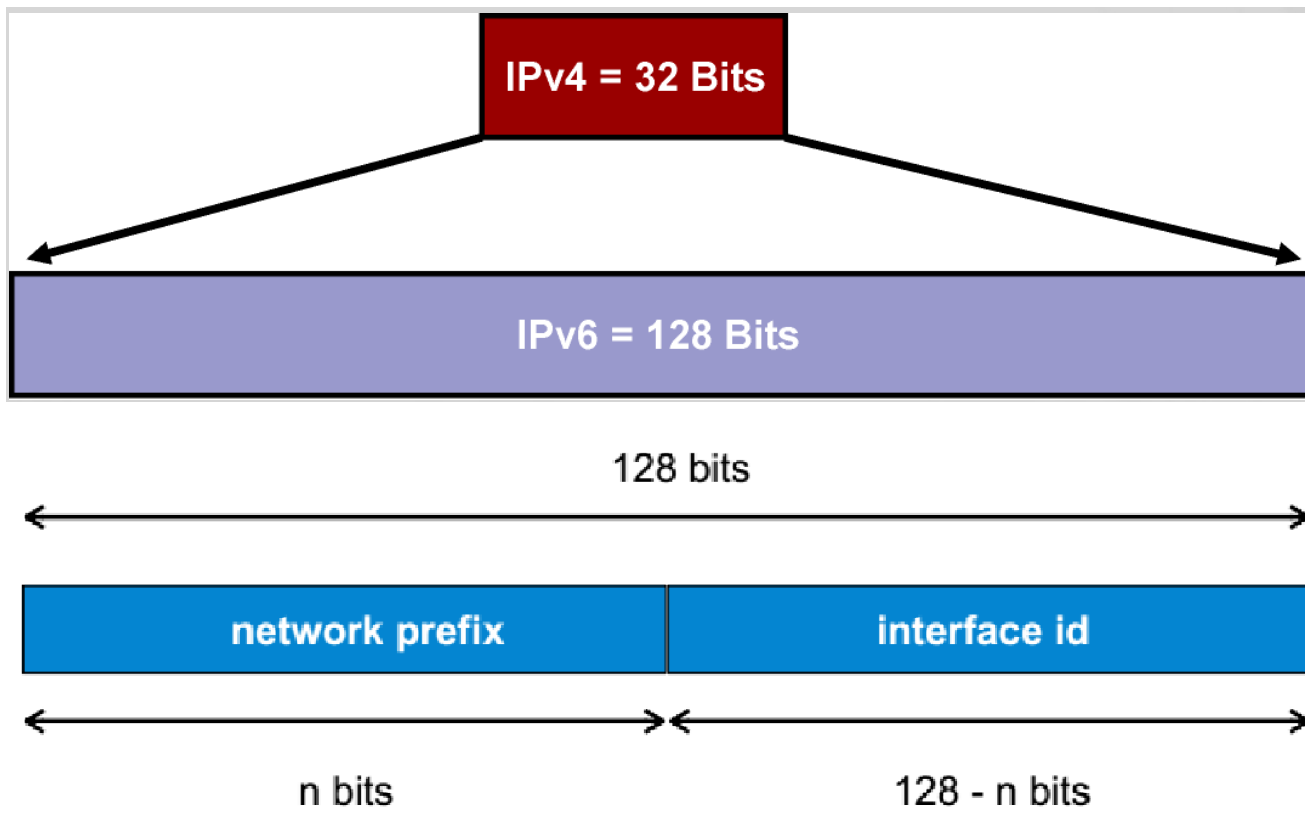
IPv6



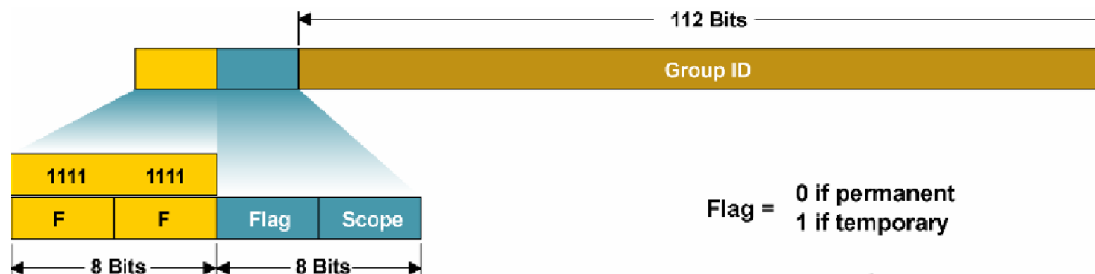
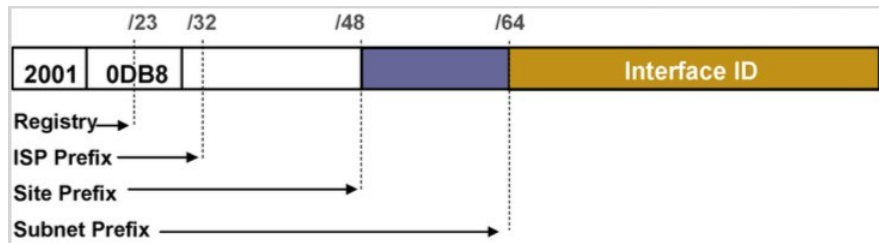
IPv6 Hlavičky



IPv6 Adresy



Typy IPv6 Adres



Flag = 0 if permanent
1 if temporary

Scope = 1 Interface-Local
2 Link-Local
3 Subnet-Local
4 Admin-Local
5 Site-Local
8 Organization
E Global

ARP

PRACTICAL NETWORKING .NET



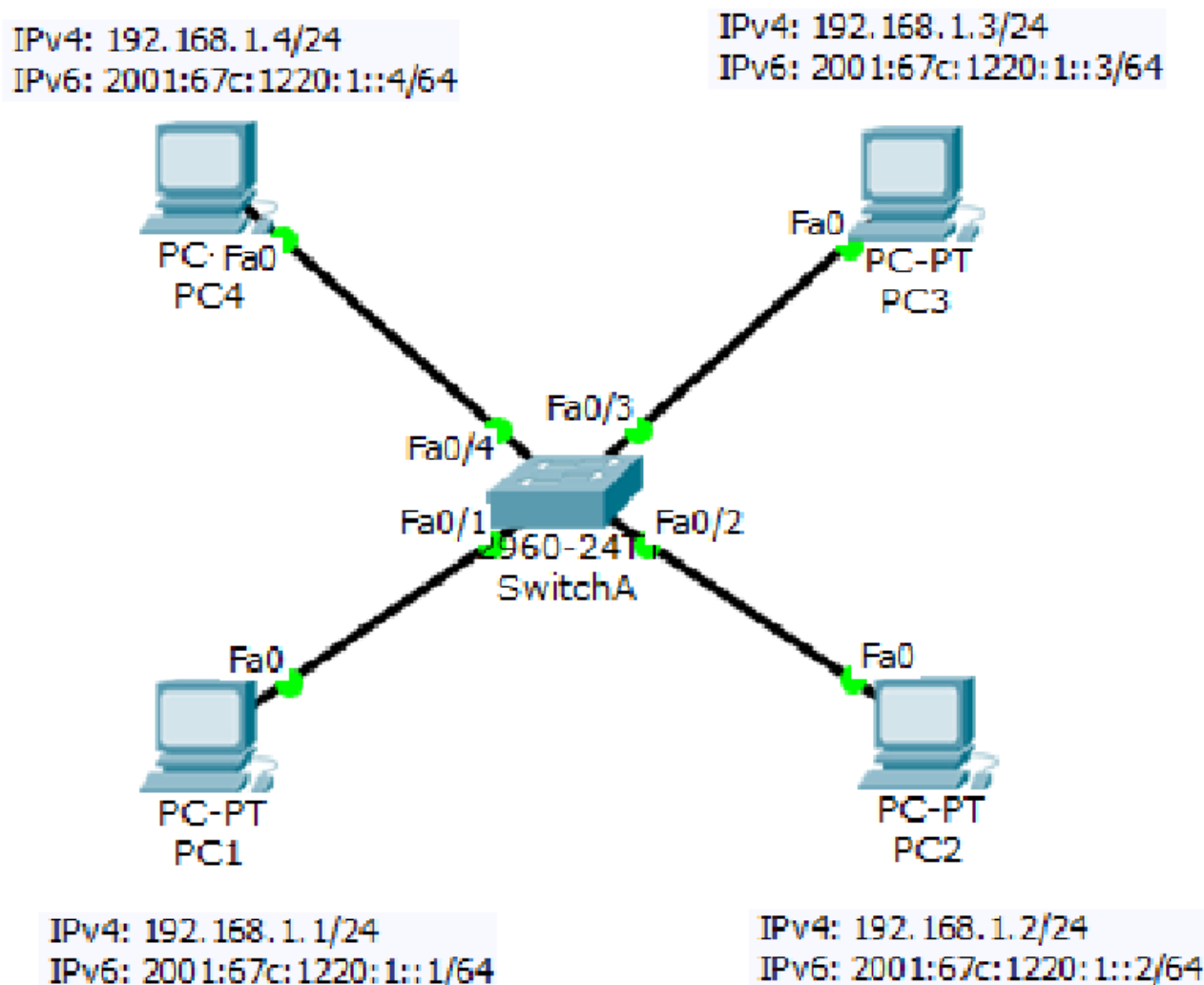
Přenos paketů



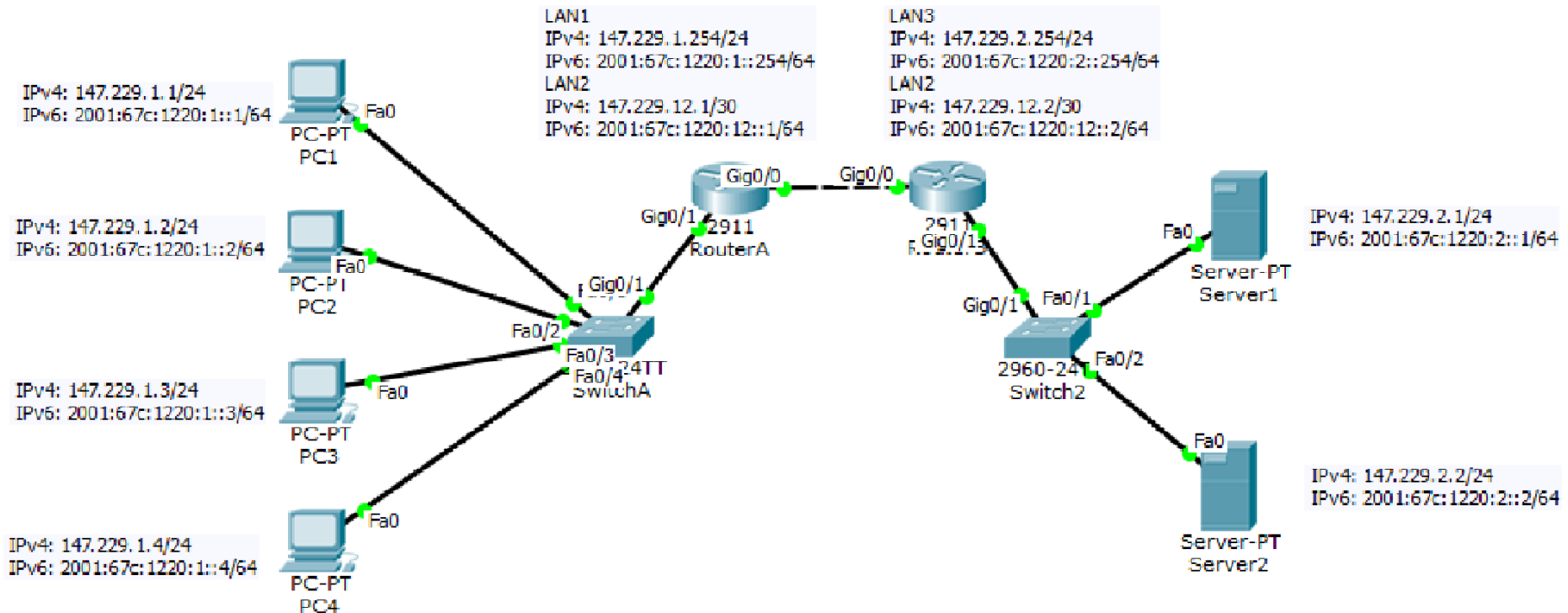
IPv4: 192.168.1.1/24
IPv6: 2001:67c:1220:1::1/64

IPv4: 192.168.1.2/24
IPv6: 2001:67c:1220:1::2/64

Přenos paketů



Přenos paketů



DHCP

DHCP server:
223.1.2.5



příchod
zí
klient



DHCP

src : 0.0.0.0,68
dest.: 255.255.255.255,67
yiaddr: 0.0.0.0
transaction ID: 654

DHCP

src: 223.1.2.5,67
dest: 255.255.255.255,68
yiaddr: 223.1.2.4
transaction ID: 654
Lifetime: 3600 secs

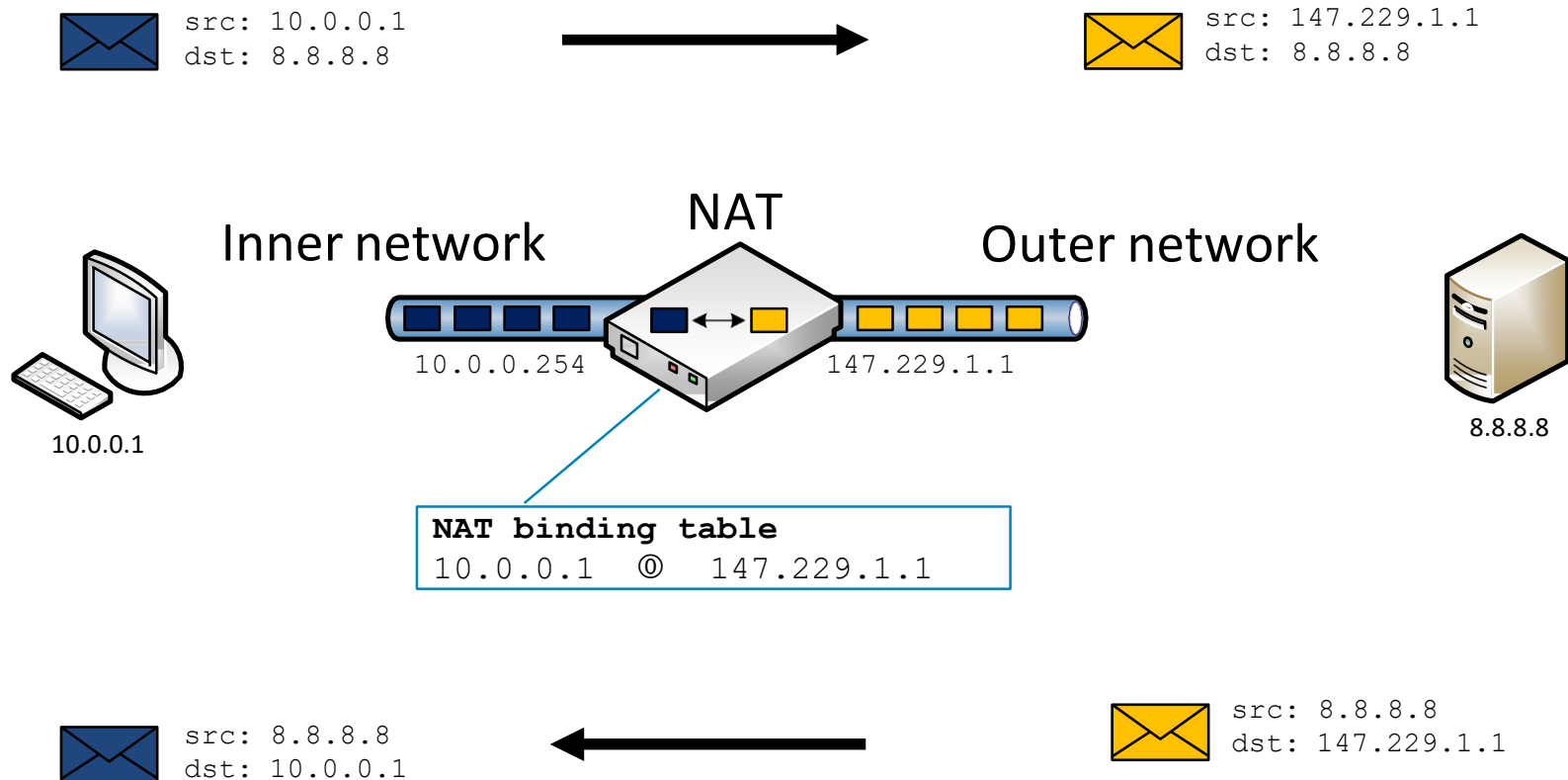
DHCP

src: 0.0.0.0,68
dest.: 255.255.255.255,67
yiaddr: 223.1.2.4
transaction ID: 655
Lifetime: 3600 secs

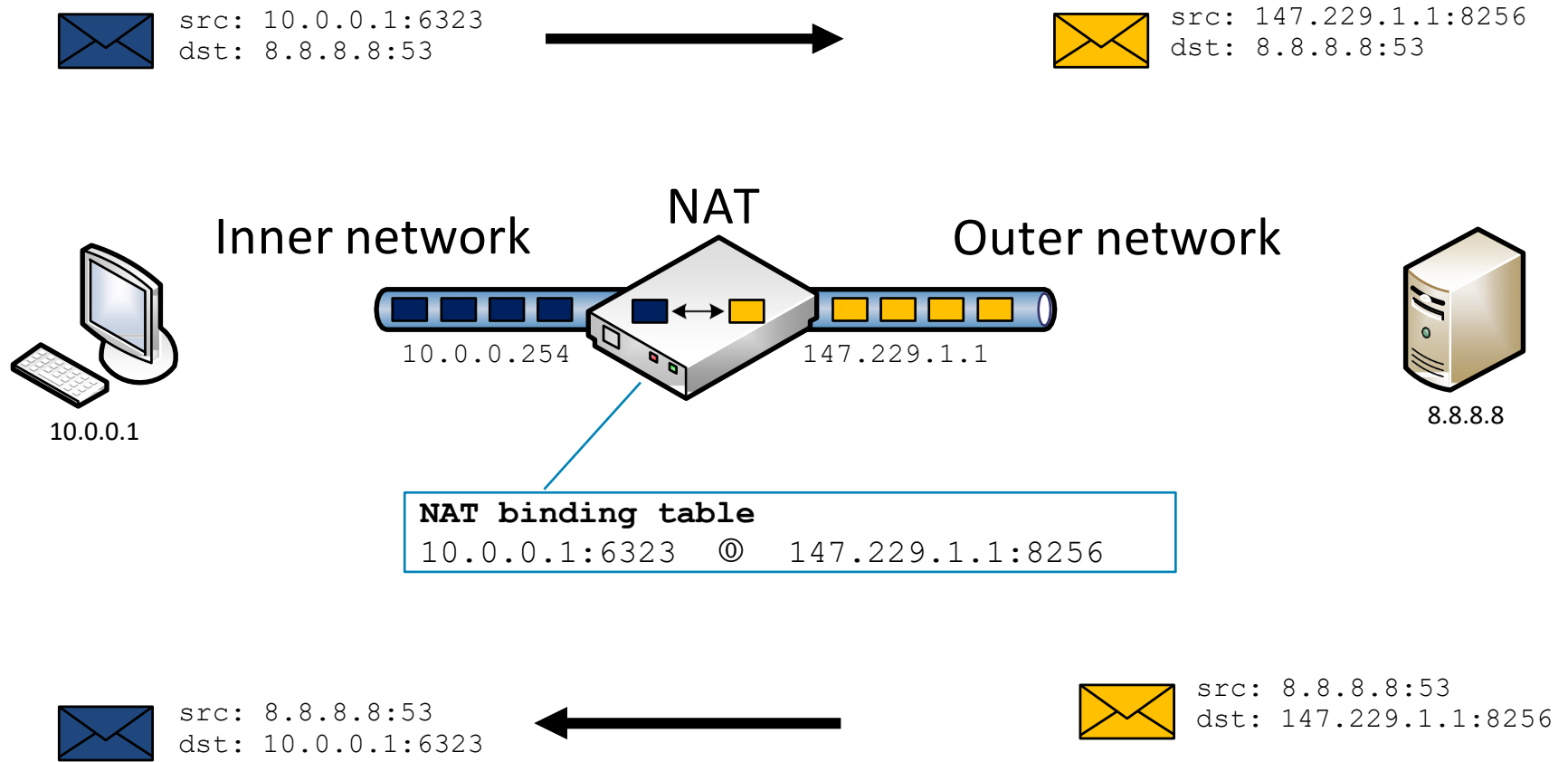
DHCP

src: 223.1.2.5,67
dest: 255.255.255.255,68
yiaddr: 223.1.2.4
transaction ID: 655
Lifetime: 3600 secs

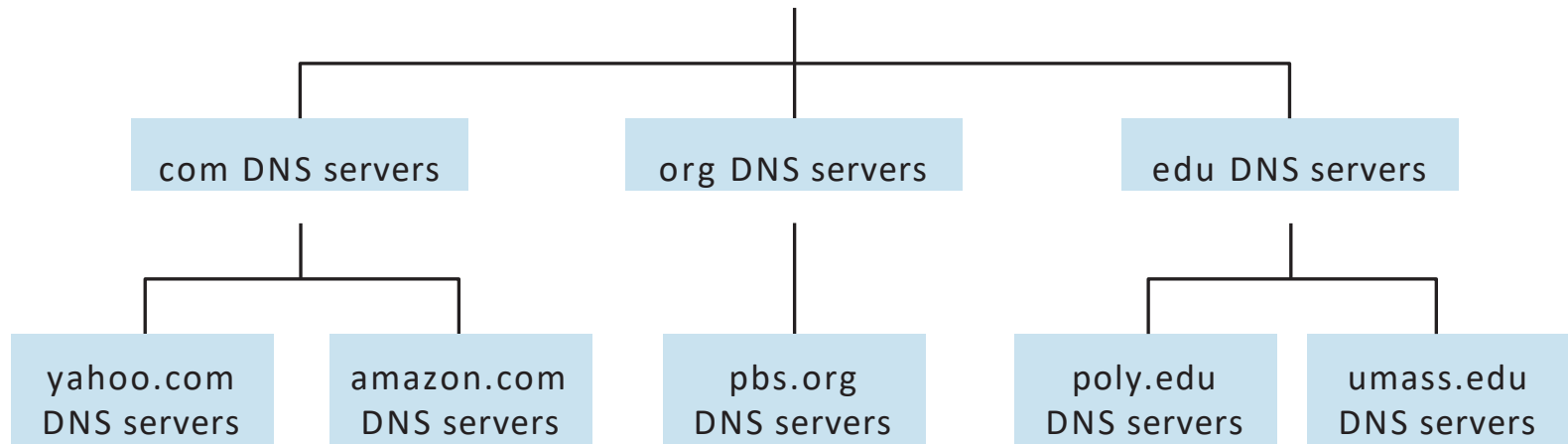
NAT



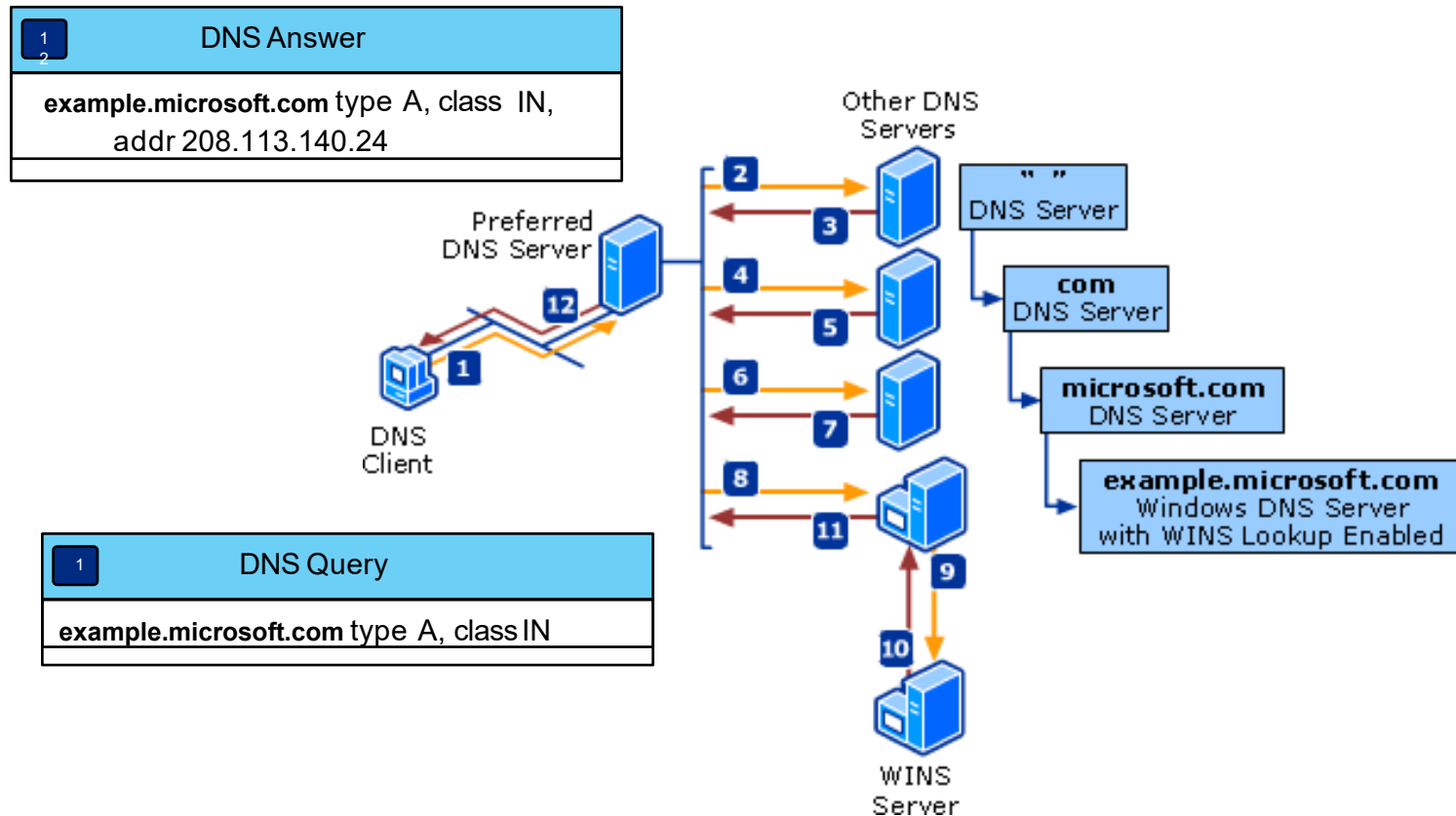
NAPT, PAT



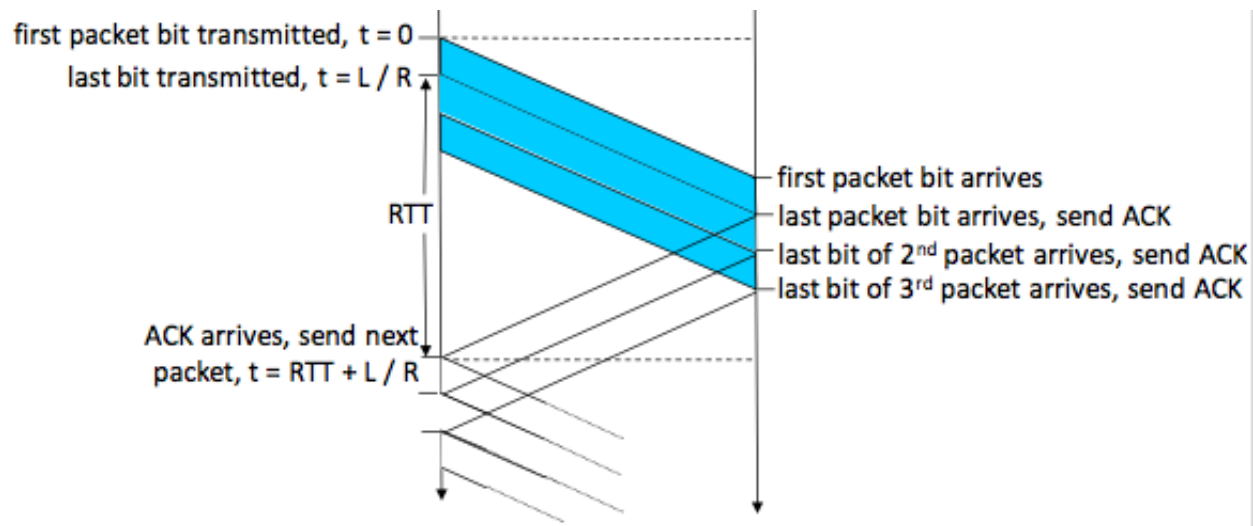
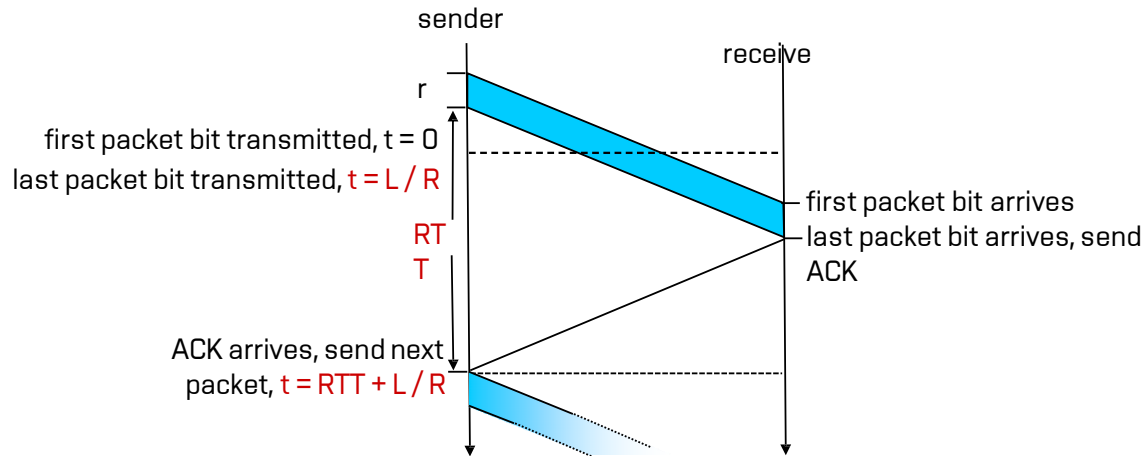
DNS: Root DNS servers



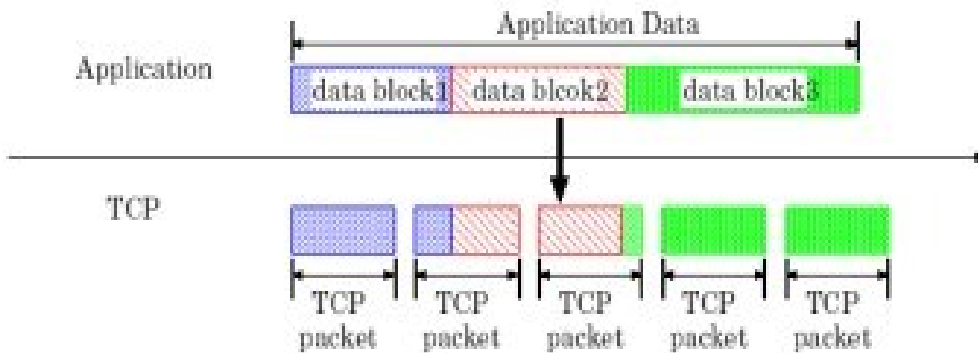
DNS: Rezoluce



Spolehlivý přenos

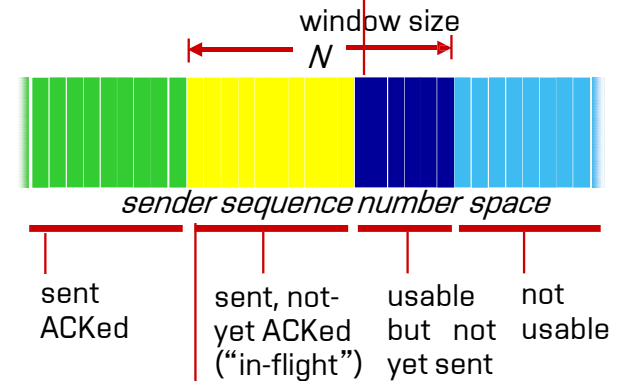


TCP



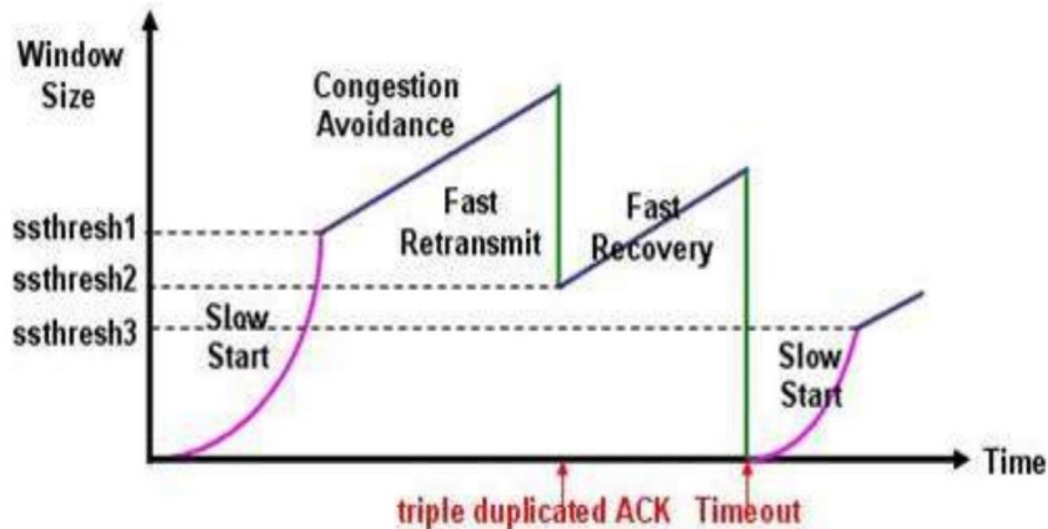
outgoing segment from sender

source port #	dest port #
sequence number	
acknowledgement number	
	rwnd
checksum	urg pointer

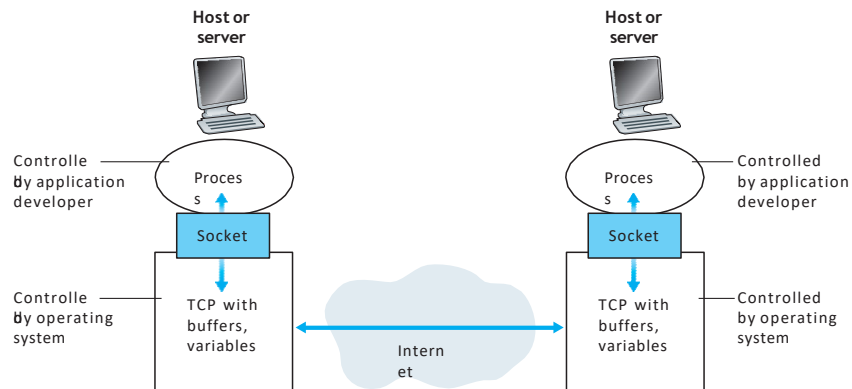


incoming segment to sender

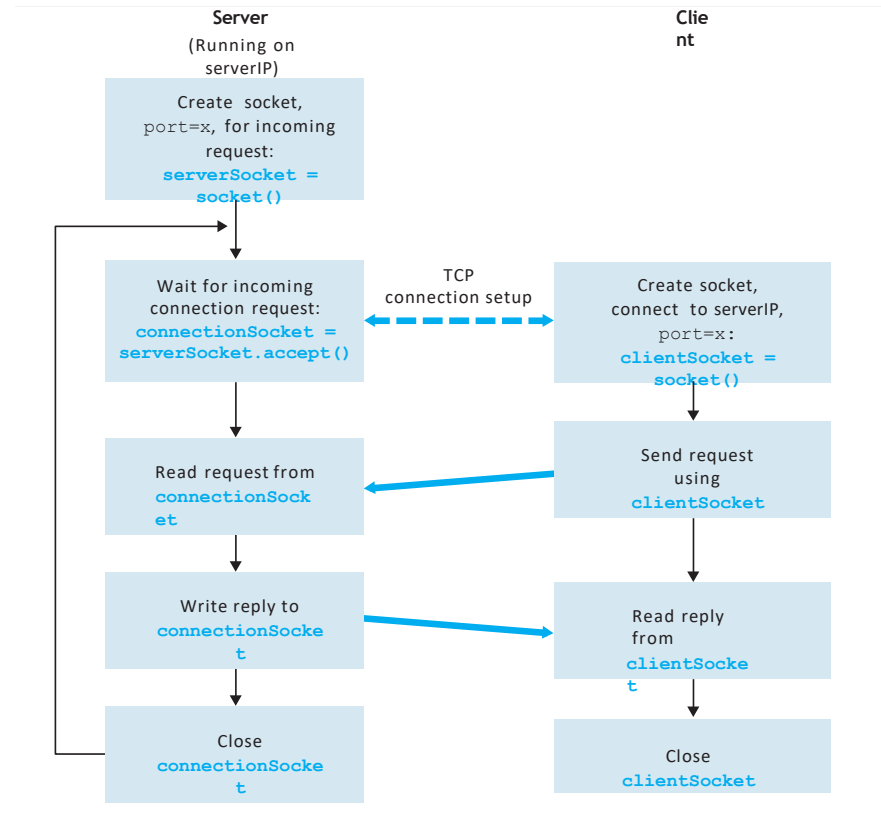
source port #	dest port #
sequence number	
acknowledgement number	
	A
	rwnd
checksum	urgpointer



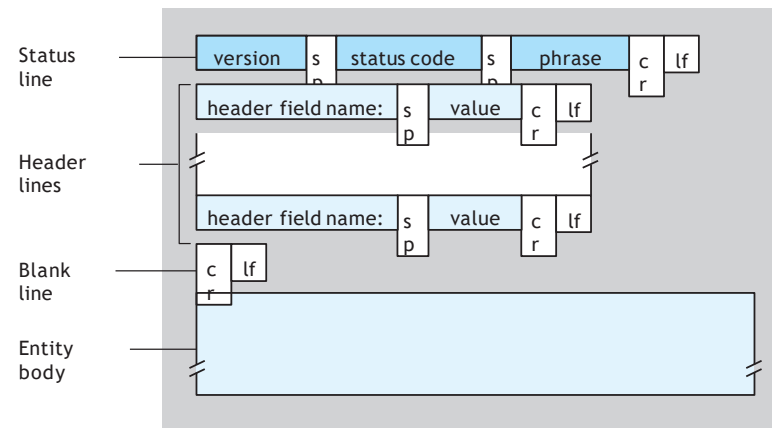
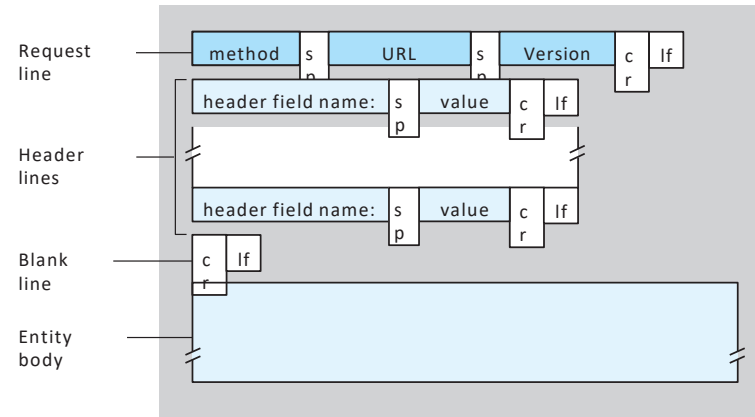
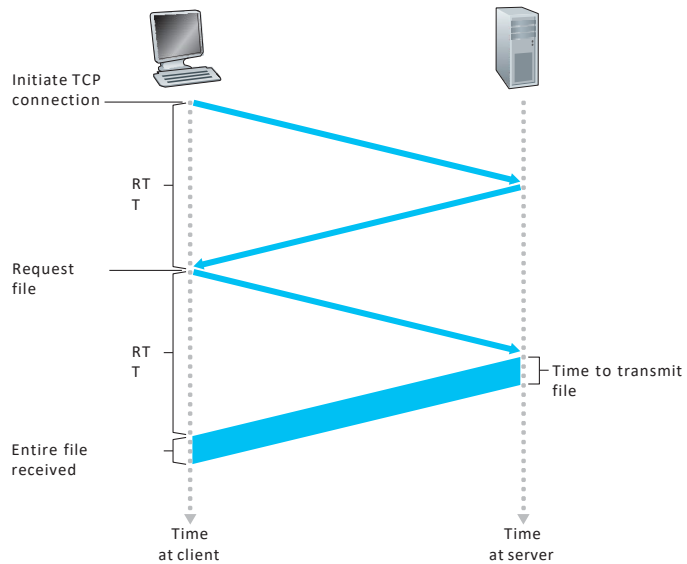
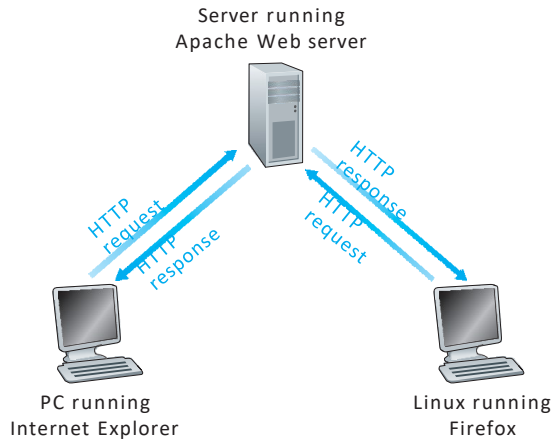
Aplikace



Socket Pair	
Socket	Socket
Address	Address
Port	Port

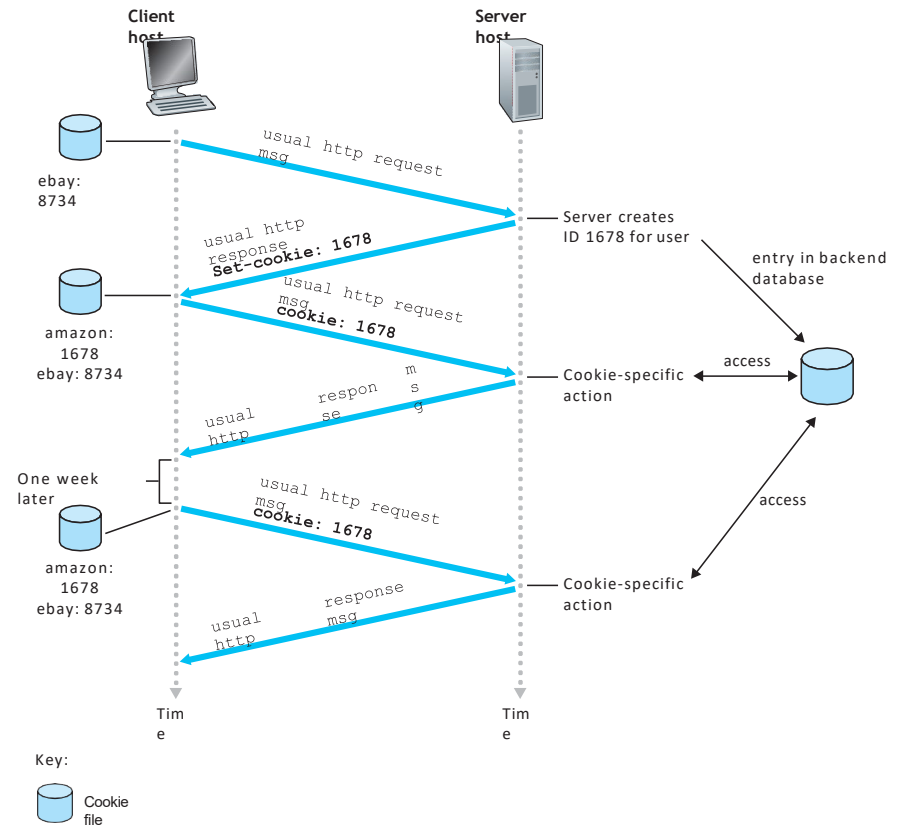
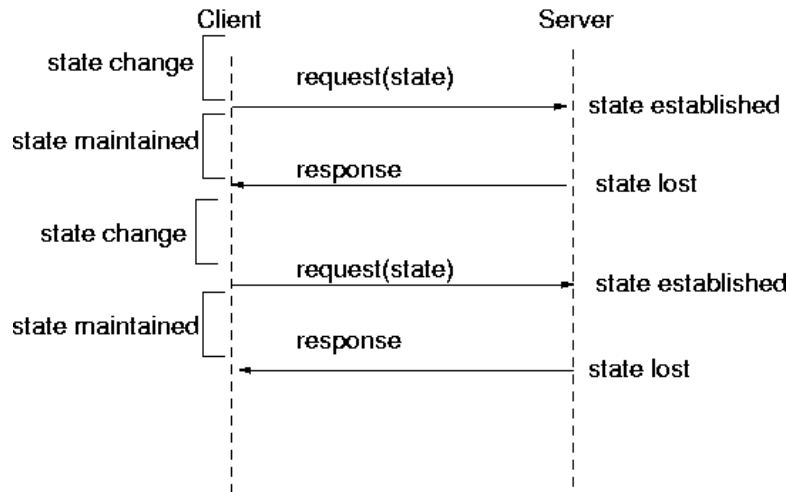


HTTP

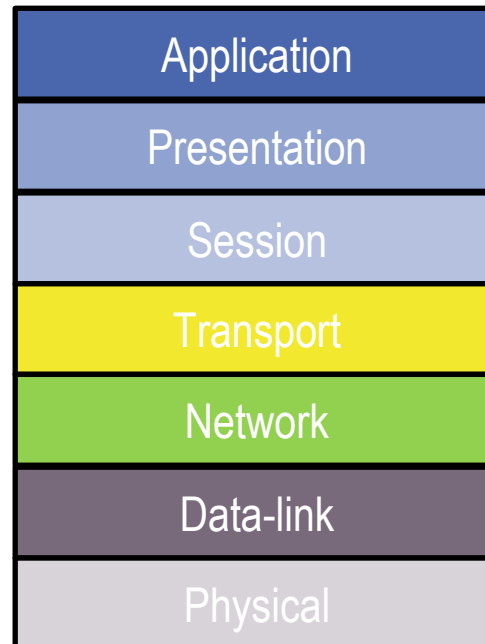


HTTPstav

Stateless server

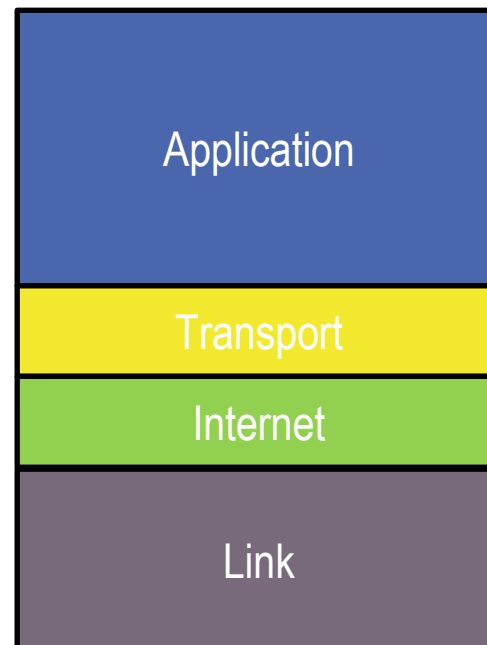


ISO-OSI RM

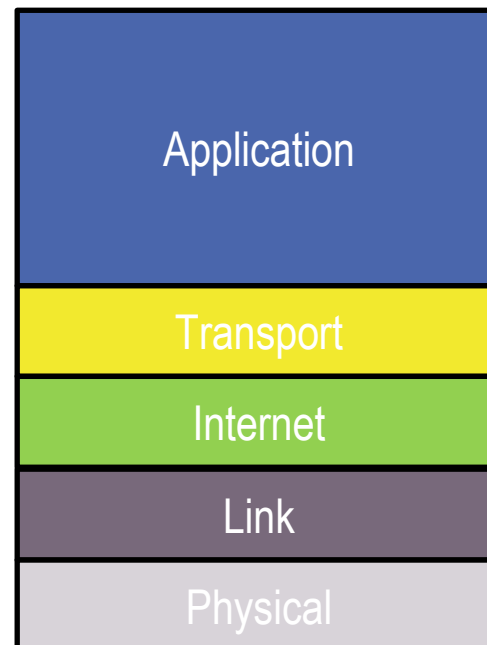


Teorie

TCP/IP RM

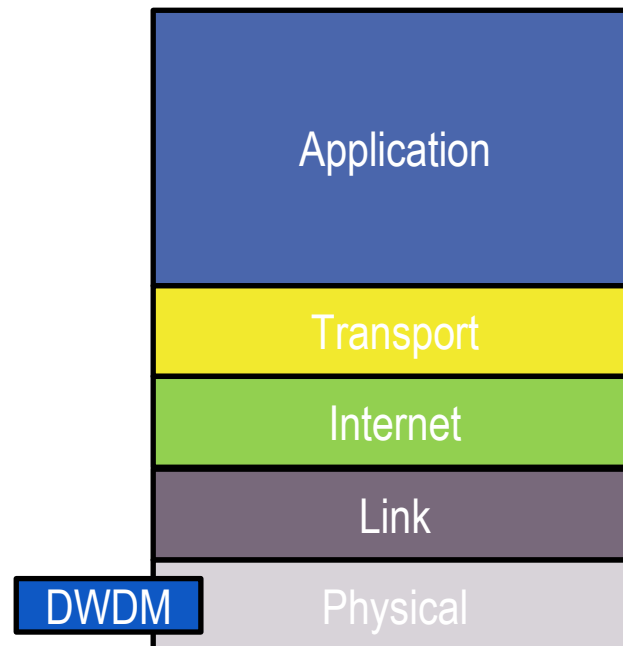


TCP/IP RM



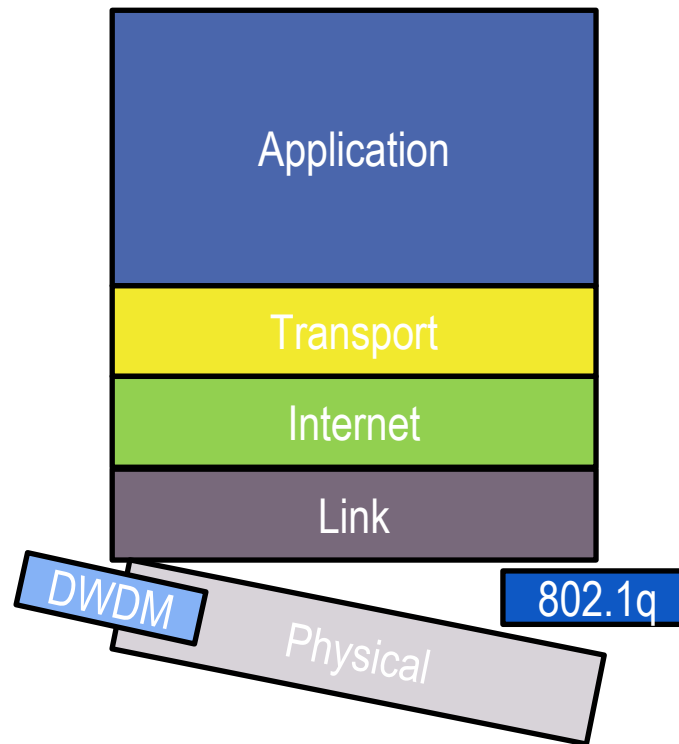
Teorie

TCP/IP RM



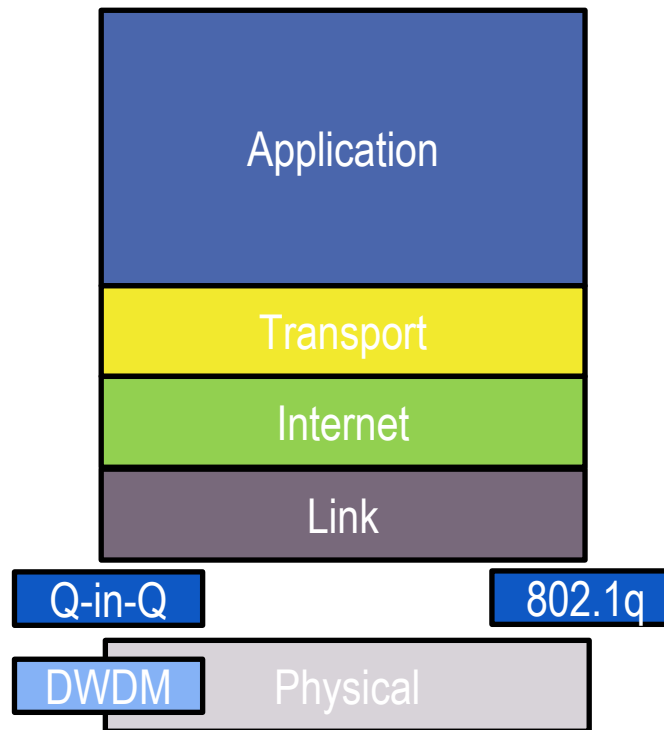
Teorie

TCP/IP RM



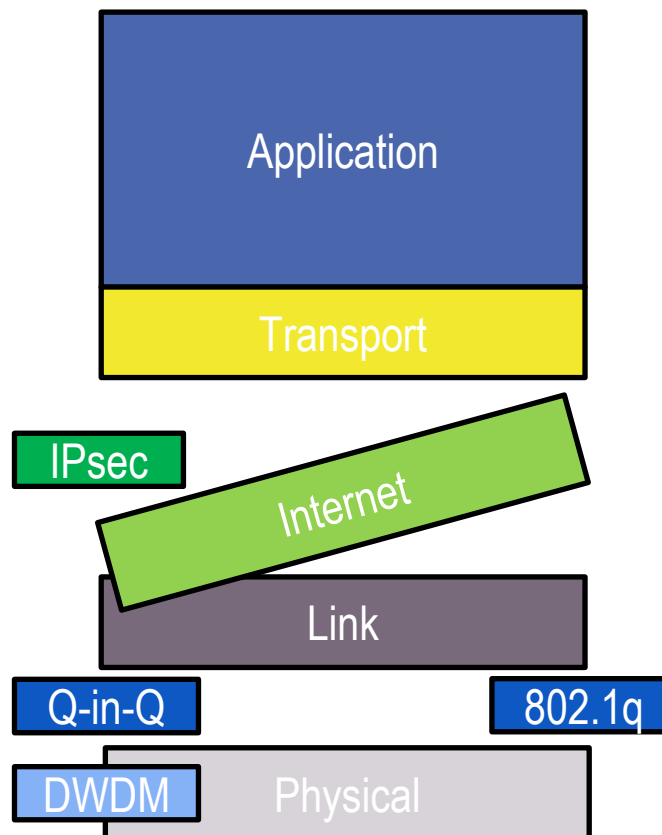
Teorie

TCP/IP RM



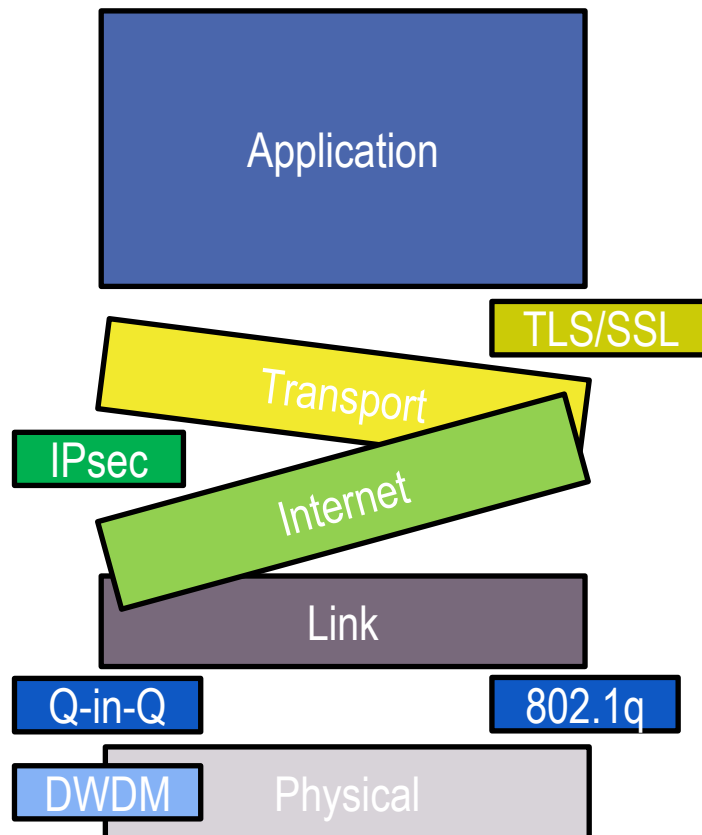
Teorie

TCP/IP RM



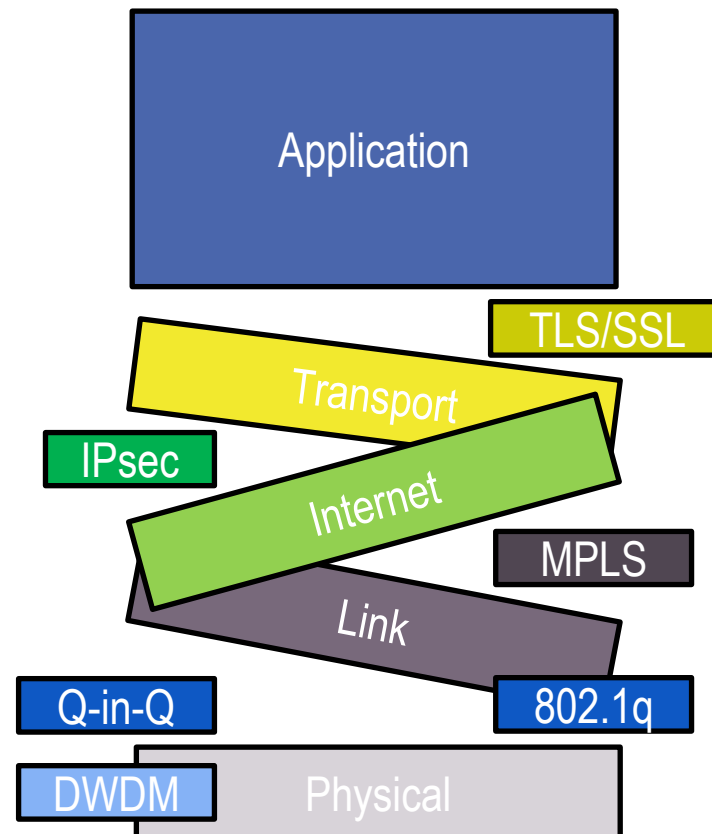
Teorie

TCP/IP RM



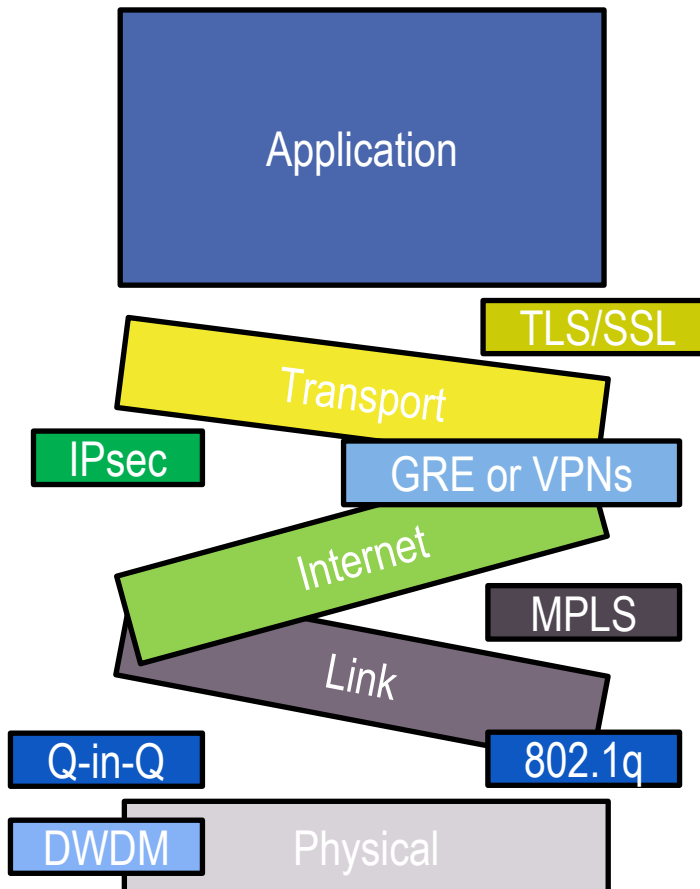
Teorie

TCP/IP RM



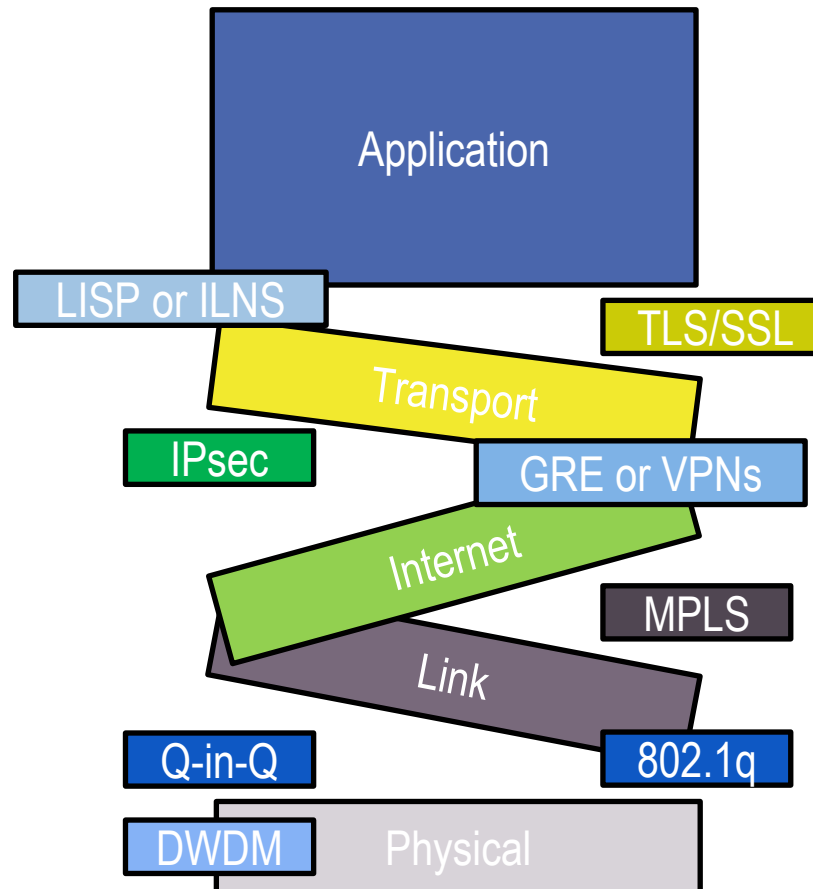
Teorie

TCP/IP RM



Teorie

TCP/IP RM



Teorie

TCP/IP RM

